

Computational Physics – Homework 2

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Summary

This assignment explored several numerical methods for solving nonlinear equations and multidimensional optimization problems. First, ordinary relaxation and overrelaxation were applied to the fixed-point equation $x = 1 - e^{-2x}$. Ordinary relaxation required 14 iterations to reach an accuracy of 10^{-6} , while a suitable choice of the overrelaxation parameter reduced the iteration count to as few as four. The numerically favorable value of ω was consistent with the local convergence analysis. The binary search method was then used to solve the nonlinear equation associated with Wien's displacement law, giving $b = 2.89777 \times 10^{-3}$ mK and an estimated solar surface temperature of approximately 5.77×10^3 K.

Finally, a multidimensional gradient-descent algorithm using numerical derivatives was implemented and first verified on the test function $f(x, y) = (x - 2)^2 + (y - 2)^2$. The same method was then used to fit a Schechter function to COSMOS galaxy stellar-mass-function data by minimizing χ^2 . The best-fit parameters were $\log_{10} \phi^* \simeq -2.5682$, $\log_{10} M^* \simeq 10.9762$, and $\alpha \simeq -1.0094$, with $\chi^2_{\min} \simeq 2.9081$. Fits starting from three distinct locations in parameter space converged to essentially the same solution, indicating that the result is robust to the tested initial conditions.

Methods

Problem 1: Relaxation and Overrelaxation

The nonlinear equation

$$x = 1 - e^{-cx}$$

was considered for the case

$$c = 2.$$

The purpose of this problem was to compare ordinary relaxation and overrelaxation, derive an error estimate for the latter, and examine how the choice of the overrelaxation parameter ω affects convergence.

(a) Error Estimate for Overrelaxation

For ordinary relaxation, the fixed-point iteration is

$$x' = f(x).$$

In overrelaxation, the update is modified to

$$x' = x + (1 + \omega)[f(x) - x],$$

or equivalently,

$$x' = (1 + \omega)f(x) - \omega x.$$

Define

$$g(x) = (1 + \omega)f(x) - \omega x.$$

The exact solution x^* is a fixed point of g , so that

$$x^* = g(x^*).$$

Let the errors before and after one iteration be

$$\epsilon = x^* - x, \quad \epsilon' = x^* - x'.$$

Expanding $g(x^*)$ to first order about x gives

$$g(x^*) \simeq g(x) + g'(x)(x^* - x).$$

Since

$$g(x^*) = x^*$$

and

$$g(x) = x',$$

we obtain

$$x^* \simeq x' + g'(x)\epsilon,$$

and hence

$$\epsilon' \simeq g'(x)\epsilon.$$

Also,

$$\epsilon - \epsilon' = (x^* - x) - (x^* - x') = x' - x,$$

so

$$\epsilon = \epsilon' + x' - x.$$

Substituting into the previous relation gives

$$\epsilon' = g'(x) (\epsilon' + x' - x).$$

Rearranging,

$$\epsilon' \simeq \frac{x - x'}{1 - \frac{1}{g'(x)}}.$$

For overrelaxation,

$$g'(x) = (1 + \omega)f'(x) - \omega.$$

Therefore, the error after one overrelaxation step is estimated by

$$\boxed{\epsilon' \simeq \frac{x - x'}{1 - \frac{1}{(1 + \omega)f'(x) - \omega}}}.$$

(b) Ordinary Relaxation

For the equation

$$x = 1 - e^{-2x},$$

the fixed-point function is

$$f(x) = 1 - e^{-2x},$$

with derivative

$$f'(x) = 2e^{-2x}.$$

The iteration

$$x_{n+1} = f(x_n)$$

was started from

$$x_0 = 1.$$

The calculation was terminated when the estimated error was smaller than

$$10^{-6}.$$

The ordinary-relaxation error estimate was obtained by taking $\omega = 0$ in the result of part (a).

(c) Overrelaxation

The same equation was then solved using

$$x_{n+1} = (1 + \omega)f(x_n) - \omega x_n.$$

The initial value was again

$$x_0 = 1,$$

and the same accuracy requirement of

$$10^{-6}$$

was used.

Several positive values of ω were tested to determine how strongly overrelaxation could reduce the number of iterations.

The local convergence factor is

$$q = (1 + \omega)f'(x^*) - \omega,$$

so near the fixed point,

$$\epsilon_{n+1} \simeq q \epsilon_n.$$

Thus, reducing the magnitude of q should improve the local convergence rate. An approximate optimal value can be estimated by setting

$$q = 0,$$

which gives

$$\omega_{\text{opt}} = \frac{f'(x^*)}{1 - f'(x^*)}.$$

(d) Negative Values of ω

The local convergence factor

$$q = (1 + \omega)f'(x^*) - \omega$$

was also used to determine whether a negative value of ω can ever improve convergence.

If

$$f'(x^*) < 0,$$

ordinary relaxation produces an alternating error: successive iterates approach the fixed point from opposite sides. A negative value of ω can reduce the magnitude of q and therefore suppress this oscillatory behavior.

Problem 2: Wien's Displacement Constant

(a) Condition for Maximum Blackbody Radiation

Planck's radiation law is

$$I(\lambda) = \frac{2\pi hc^2 \lambda^{-5}}{e^{hc/(\lambda k_B T)} - 1}.$$

Defining

$$a = \frac{hc}{k_B T},$$

the logarithm of the intensity can be written as

$$\ln I = \text{constant} - 5 \ln \lambda - \ln(e^{a/\lambda} - 1).$$

At the wavelength of maximum intensity,

$$\frac{dI}{d\lambda} = 0,$$

which gives

$$-\frac{5}{\lambda} + \frac{ae^{a/\lambda}}{\lambda^2(e^{a/\lambda} - 1)} = 0.$$

Introducing

$$x = \frac{hc}{\lambda k_B T}$$

reduces this equation to

$$5e^{-x} + x - 5 = 0.$$

Since

$$x = \frac{hc}{\lambda k_B T},$$

the wavelength of maximum radiation satisfies

$$\lambda T = \frac{hc}{k_B x}.$$

Thus Wien's displacement law is

$$\lambda_{\max} = \frac{b}{T},$$

where

$$b = \frac{hc}{k_B x}.$$

(b) Binary Search Solution

The nonlinear equation

$$f(x) = 5e^{-x} + x - 5 = 0$$

was solved using the binary search method.

The nonzero positive root was bracketed in the interval

$$4 < x < 5.$$

At each iteration, the interval was divided in half and the subinterval containing the sign change was retained. The search was stopped when the interval width became smaller than

$$10^{-6}.$$

The resulting root was then used to calculate the Wien displacement constant,

$$b = \frac{hc}{k_B x}.$$

(c) Solar Surface Temperature

Using the measured peak wavelength of the solar spectrum,

$$\lambda_{\max} = 502 \text{ nm},$$

the solar surface temperature was estimated using

$$T = \frac{b}{\lambda_{\max}}.$$

Problem 3: Multidimensional Gradient Descent

A multidimensional gradient-descent algorithm was implemented using numerical derivatives. For a parameter vector $\boldsymbol{\theta}$, each component of the gradient was approximated using the central-difference formula

$$\frac{\partial f}{\partial \theta_j} \approx \frac{f(\boldsymbol{\theta} + h_j \hat{e}_j) - f(\boldsymbol{\theta} - h_j \hat{e}_j)}{2h_j}.$$

The parameters were updated according to

$$\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n - \eta \nabla f,$$

where η is the gradient-descent step size. A backtracking procedure was used to reduce the step size whenever a proposed update increased the objective function.

Test Function

The algorithm was first tested on

$$f(x, y) = (x - 2)^2 + (y - 2)^2,$$

whose exact minimum is known to be

$$(x, y) = (2, 2).$$

Only numerical derivatives were used, even though the analytical derivatives of the test function are known.

Schechter-Function Fit

The COSMOS stellar mass-function measurements were modeled using

$$n(M_{\text{gal}}) = \phi^* \left(\frac{M_{\text{gal}}}{M^*} \right)^{\alpha+1} \exp \left(-\frac{M_{\text{gal}}}{M^*} \right) \ln(10).$$

To improve numerical stability, the optimized parameter vector was chosen as

$$\boldsymbol{\theta} = (\log_{10} \phi^*, \log_{10} M^*, \alpha).$$

For each trial parameter set, the goodness of fit was quantified using

$$\chi^2 = \sum_i \left[\frac{n_i - n_{\text{model}}(M_i)}{\sigma_i} \right]^2.$$

The numerical-gradient descent algorithm was applied to minimize χ^2 . To test the robustness of the solution, the optimization was repeated from three distinct initial parameter vectors,

$$(-2.5, 10.8, -1.2),$$

$$(-3.0, 11.0, -1.0),$$

and

$$(-2.8, 11.2, -0.8).$$

The convergence history of χ^2 was recorded for each run, and the final best-fit Schechter function was compared directly with the COSMOS data on a log-log plot.

Results

Problem 1: Relaxation and Overrelaxation

(a) Error Estimate for Overrelaxation

The derivation gives the following estimate for the error after one overrelaxation step:

$$\epsilon' \simeq \frac{x - x'}{1 - \frac{1}{(1 + \omega)f'(x) - \omega}}.$$

For $\omega = 0$, this reduces to the corresponding error estimate for ordinary relaxation.

(b) Ordinary Relaxation

For $c = 2$, ordinary relaxation converged to the nonzero solution

$$x^* \approx 0.79681263.$$

Starting from

$$x_0 = 1,$$

the method required

$$\boxed{14 \text{ iterations}}$$

to reach the required accuracy of 10^{-6} .

The final numerical value was

$$x = 0.7968126311,$$

with an estimated error of

$$5.01 \times 10^{-7}.$$

(c) Overrelaxation

Overrelaxation significantly reduced the number of iterations required for convergence. The numerical results for several choices of ω are shown in Table 1.

ω	Solution x	Iterations	Estimated error
0.00	0.7968126311	14	5.01×10^{-7}
0.10	0.7968125600	12	4.30×10^{-7}
0.20	0.7968126298	10	5.00×10^{-7}
0.30	0.7968129418	8	8.12×10^{-7}
0.40	0.7968124315	7	3.02×10^{-7}
0.50	0.7968123730	4	2.43×10^{-7}
0.60	0.7968120294	5	1.01×10^{-7}
0.65	0.7968118338	4	2.96×10^{-7}
0.68	0.7968121261	4	3.88×10^{-9}
0.70	0.7968121085	4	2.16×10^{-8}
0.75	0.7968128656	4	7.35×10^{-7}

Table 1: Convergence of the overrelaxation method for several values of ω . Ordinary relaxation corresponds to $\omega = 0$.

For example, using

$$\omega = 0.4$$

reduced the iteration count from 14 to 7, approximately a factor of two. Values in the range

$$\omega \approx 0.5\text{--}0.7$$

converged even faster, typically requiring only 4–5 iterations.

At the fixed point,

$$f'(x^*) = 2e^{-2x^*} \approx 0.406375.$$

The approximate locally optimal value is therefore

$$\omega_{\text{opt}} = \frac{f'(x^*)}{1 - f'(x^*)} \approx 0.6846.$$

This agrees well with the numerical results. For

$$\omega = 0.68,$$

the algorithm converged in four iterations with an estimated error of

$$3.88 \times 10^{-9}.$$

Thus, a suitable choice of ω can make overrelaxation substantially faster than ordinary relaxation.

(d) Negative Values of ω

A negative value of ω can be useful when

$$f'(x^*) < 0.$$

In this case, ordinary relaxation approaches the solution by oscillating from one side of the fixed point to the other because the error changes sign after each iteration.

The effective convergence factor for overrelaxation is

$$q = (1 + \omega)f'(x^*) - \omega.$$

If $f'(x^*) < 0$, an appropriately chosen negative value of ω can reduce

$$|q|,$$

thereby reducing the oscillation and causing the error to decay more rapidly.

Therefore, although positive ω accelerates convergence for the equation studied in part (c), negative ω can be advantageous for fixed-point problems whose iteration function has a negative slope at the solution.

Problem 2: Wien's Displacement Constant

(a) Maximum-Radiation Condition

Differentiating Planck's radiation law and introducing

$$x = \frac{hc}{\lambda k_B T}$$

gives the nonlinear equation

$$\boxed{5e^{-x} + x - 5 = 0}$$

for the wavelength at which the blackbody radiation intensity is maximized.

Since

$$x = \frac{hc}{\lambda k_B T},$$

the wavelength of maximum radiation satisfies

$$\lambda_{\max} T = \frac{hc}{k_B x}.$$

Therefore, defining

$$b = \frac{hc}{k_B x},$$

gives Wien's displacement law,

$$\boxed{\lambda_{\max} = \frac{b}{T}}.$$

(b) Wien Displacement Constant

The nonzero positive root of

$$5e^{-x} + x - 5 = 0$$

was found using binary search on the interval

$$4 < x < 5.$$

Using a target accuracy of

$$10^{-6},$$

the method converged in

$$\boxed{20 \text{ iterations}}$$

to

$$x = 4.9651141167.$$

The residual at the numerical solution was

$$f(x) \approx -1.11 \times 10^{-7}.$$

Using this root in

$$b = \frac{hc}{k_B x}$$

gave

$$\boxed{b = 2.8977720223 \times 10^{-3} \text{ m K}}.$$

(c) Solar Surface Temperature

The wavelength of maximum solar radiation was taken to be

$$\lambda_{\max} = 502 \text{ nm} = 5.020 \times 10^{-7} \text{ m}.$$

Using Wien's displacement law,

$$T = \frac{b}{\lambda_{\max}},$$

gives

$$T = \frac{2.8977720223 \times 10^{-3}}{5.020 \times 10^{-7}} \approx 5772.45 \text{ K}.$$

Thus the estimated surface temperature of the Sun is

$$\boxed{T \approx 5.77 \times 10^3 \text{ K}}.$$

Problem 3: Gradient Descent and Schechter-Function Fitting

The numerical-gradient descent algorithm was first tested on

$$f(x, y) = (x - 2)^2 + (y - 2)^2.$$

The algorithm successfully converged to the known minimum at

$$(x, y) = (2, 2),$$

confirming that the multidimensional gradient-descent implementation was working correctly.

The same algorithm was then applied to the COSMOS stellar mass-function data. The parameters were optimized in the form

$$(\log_{10} \phi^*, \log_{10} M^*, \alpha),$$

and the objective function was the chi-square,

$$\chi^2 = \sum_i \left[\frac{n_i - n_{\text{model}}(M_i)}{\sigma_i} \right]^2.$$

Starting from three distinct locations in parameter space, the algorithm converged to nearly identical solutions. The fitted parameters were approximately

$$\log_{10} \phi^* = -2.5682$$

$$\log_{10} M^* = 10.9762$$

and

$$\alpha = -1.0094.$$

These correspond to

$$\phi^* \approx 2.70 \times 10^{-3},$$

and

$$M^* \approx 9.47 \times 10^{10}.$$

The minimum chi-square was

$$\chi_{\min}^2 \approx 2.9081.$$

The three starting points required 156, 217, and 164 iterations, respectively, but converged to the same minimum within numerical precision. This demonstrates that the fitted solution is robust to reasonable changes in the initial parameter values.

Figure 1 shows the gradient-descent path for the test function, while Figure 2 shows the decrease of χ^2 during the Schechter fit for the three initial guesses. Figure 3 compares the best-fit Schechter function with the COSMOS data.

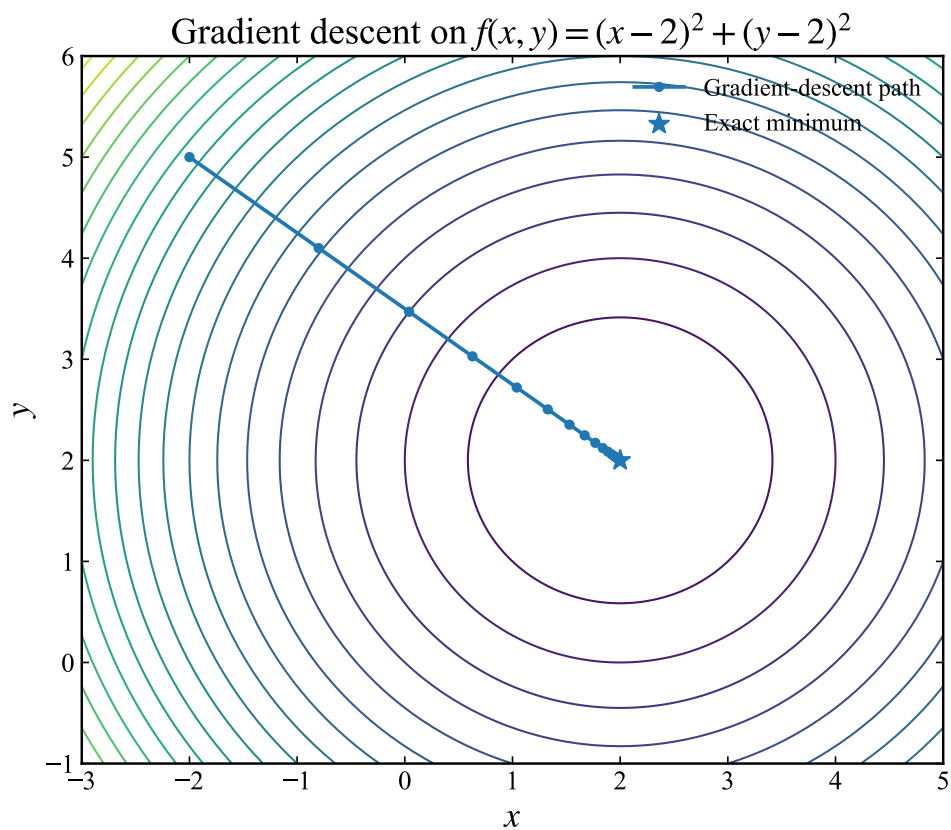


Figure 1: Gradient-descent path for the test function $f(x, y) = (x - 2)^2 + (y - 2)^2$. The numerical method converges to the known minimum at $(2, 2)$.

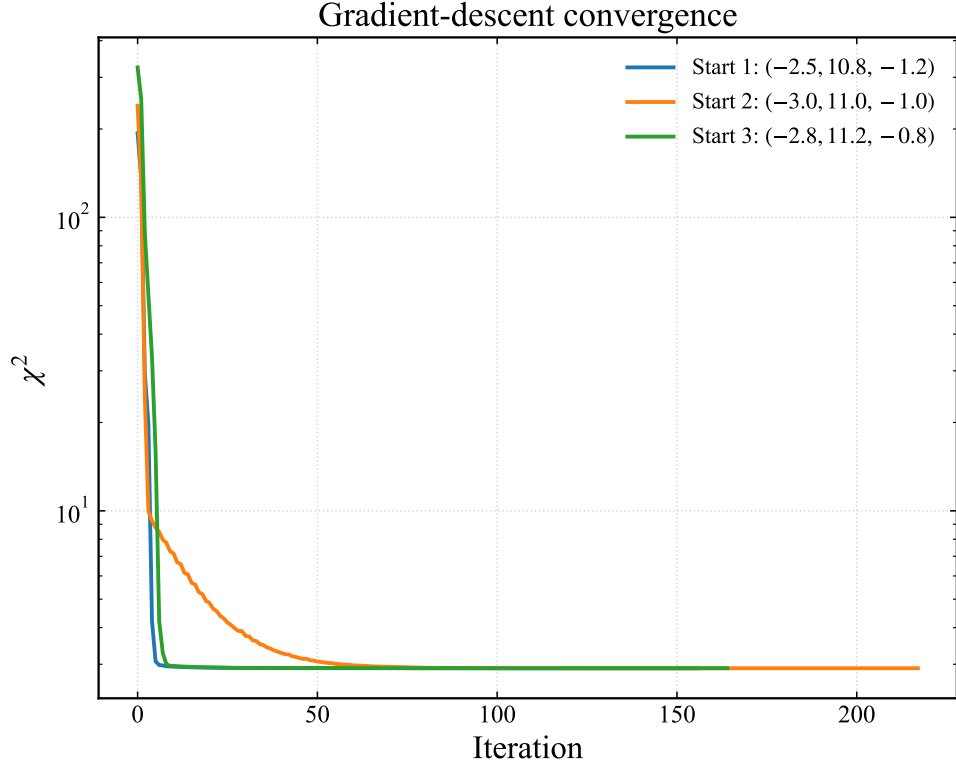


Figure 2: Evolution of χ^2 with iteration number for three distinct initial guesses in Schechter-parameter space. All three runs converge to the same minimum, demonstrating robustness to the initial conditions.

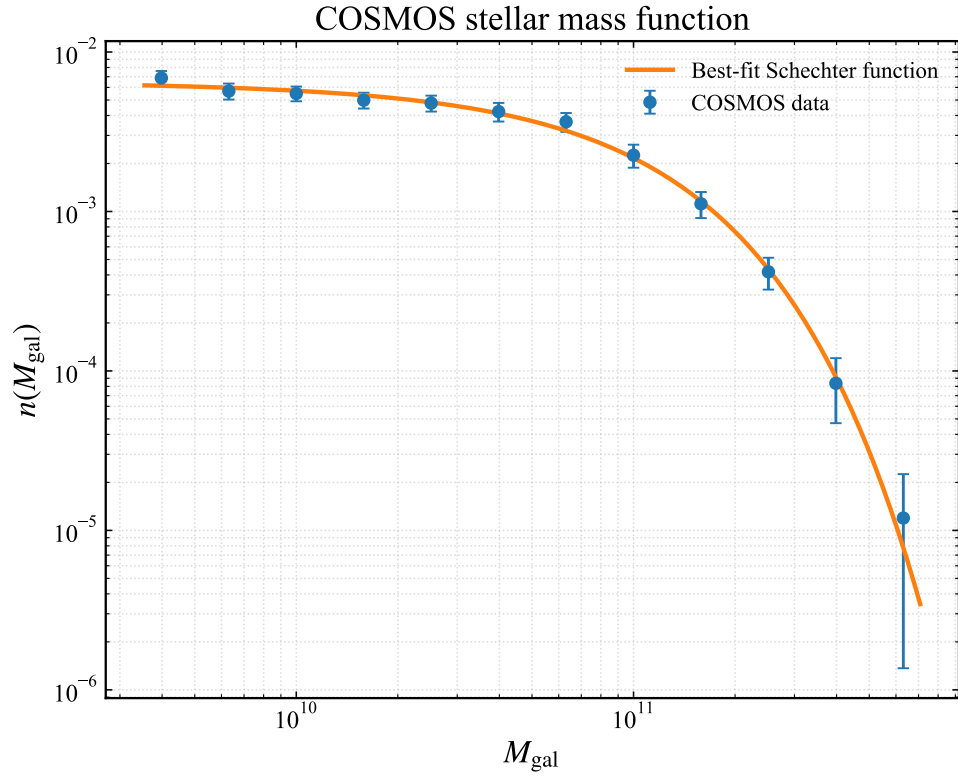


Figure 3: COSMOS galaxy stellar mass-function measurements compared with the best-fit Schechter function. Both axes are shown on logarithmic scales.

Code

The Python code used for this assignment is available in the `homework2` directory of:

<https://github.com/chengxuanSONG/computational-physics>